

Question 1 (15 Marks)

- (a) If $z = 1 + \sqrt{3}i$ and $\omega = -1 - i$, find
- (i) $|z|$ 1
 - (ii) $\arg \omega$ 1
 - (iii) $\omega \bar{z}$ 2
- (b) (i) Find the square roots of $-3 - 4i$ 2
- (ii) Hence, solve $z^2 - 3z + (3 + i) = 0$ 2
- (c) Suppose $P(x) = x^3 - 3x^2 + 4x - 2$
- (i) Show that $P(1 - i) = 0$ 1
 - (ii) Explain why $x = 1 + i$ is also a root of $P(x)$ 1
 - (iii) Hence write $P(x)$ as a product of a linear and quadratic factor over $\mathbb{R}$ 2
- (d) Sketch the region represented by the following inequations: 3
- $$-\frac{\pi}{4} < \arg z < \frac{\pi}{4} \text{ and } \operatorname{Re}(z) \leq 2$$

Question 2 (10 Marks)

- (a) For the following statement:
- “For every integer n , if n is divisible by 3 then n^2 is divisible by 3.”
- Write down its
- (i) Converse. 1
 - (ii) Contrapositive 1
 - (iii) Negation 1
- (b) Prove that the following is true for all positive integers n : n is even if and only if $3n^2 + 8$ is even 4
- (c) Using a proof by contradiction, prove that the equation $2x^3 + 6x + 1 = 0$ has no integer solutions. 3

Question 3 (16 marks)

- (a) The complex number z is represented by the point C in the Argand diagram. Find and sketch the curve that C is on if 4

$$\operatorname{Re}\left(\frac{z-1}{z-i}\right) = 0$$

- (b) The points A and C represent the complex number $z_1 = 1 + i$ and $z_3 = 7 + 3i$ respectively and B represents the complex number z_2 . Find the complex number(s) z_2 in the form of $a + ib$ where a and b are real such that $\triangle ABC$ is an isosceles triangle that is right angled at B . 4

- (c) Consider the equation $z^7 = 1$

- (i) Show that the roots of the equation are given by 2

$$\alpha_k = \cos\left(\frac{2k\pi}{7}\right) + i \sin\left(\frac{2k\pi}{7}\right), k = 1, 2, 3, 4, 5, 6, 7$$

- (ii) By using the factorisation $z^7 - 1 = (z - 1)(z^6 + z^5 + z^4 + z^3 + z^2 + z + 1)$, or otherwise 2

$$\text{Show } (1 - \alpha_1)(1 - \alpha_2)(1 - \alpha_3) \dots (1 - \alpha_6) = 7$$

- (iii) Show that $1 - \alpha_k = 2 \sin\left(\frac{k\pi}{7}\right) \left[\cos\left(\frac{\pi}{14}(2k - 7)\right) + i \sin\left(\frac{\pi}{14}(2k - 7)\right) \right]$ 2

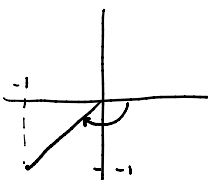
- (iv) Hence show that 2

$$\sin\left(\frac{\pi}{7}\right) \sin\left(\frac{2\pi}{7}\right) \sin\left(\frac{3\pi}{7}\right) \dots \sin\left(\frac{6\pi}{7}\right) = \frac{7}{2^6}$$

Question 1

a) i) $|z| = \sqrt{1^2 + (\sqrt{3})^2}$
 $= \sqrt{4}$
 $= 2$

iii) $w\bar{z} = (-1-i)(1-\sqrt{3}i)$ [1]
 $= -1 + \sqrt{3}i - i - \sqrt{3}$
 $= (-1-\sqrt{3}) + i(\sqrt{3}-1)$ [1]

ii) 
$$\arg(w) = -\pi + \tan^{-1}(1)$$
$$= -\pi + \frac{\pi}{4}$$
$$= -\frac{3\pi}{4}$$

b) i) let $w^2 = -3-4i$
where $w = a+ib$

$$\therefore a^2 - b^2 = -3$$

$$2ab = -4$$

$$ab = -2$$

$\therefore$ by inspection

$$a = 1 \quad b = -2 \quad \text{or} \quad a = -1 \quad b = 2$$

$$\therefore w = 1-2i \quad \text{or} \quad w = -1+2i$$

ii) $z^2 - 3z + (3+i) = 0$ $\Delta = 9 - 4 \times 1 \times (3+i)$
 $= 9 - 12 - 4i$
 $= -3 - 4i$

$$\therefore z = \frac{3+w}{2} \quad (\text{from i})$$

$$\therefore z = \frac{1}{2}(3 + (1-2i)) \quad \text{or} \quad \frac{1}{2}(3 + (-1+2i))$$
$$= 2-i \quad \text{or} \quad 1+i$$

c) i) $p(x) = x^3 - 3x^2 + 4x - 2$

$$p(1-i) = (1-i)^3 - 3(1-i)^2 + 4(1-i) - 2$$
$$= (-2i)(1-i) - 3(-2i) + 4 - 4i - 2$$
$$= -2i - 2 + 6i + 4 - 4i - 2$$
$$= 0$$

$\therefore x = 1-i$ is a zero of $p(x)$

$$(1-i)(1-i) = 1 - 2i + i^2$$
$$= 1 - 2i - 1$$
$$= -2i$$

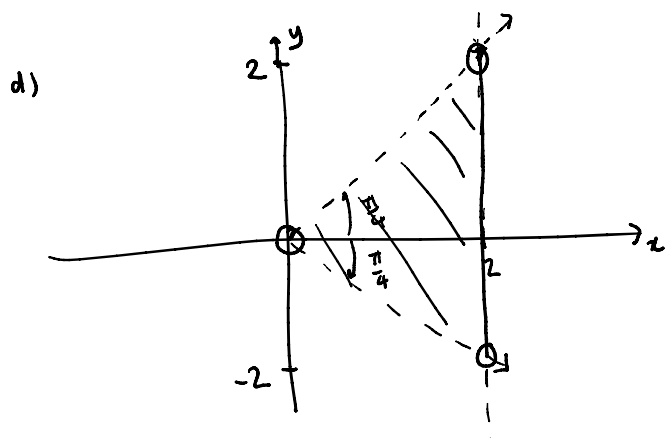
ii) As the coefficients of all terms in $p(x)$ are real
then by the conjugate root theorem $x = 1+i$ is also a zero

$$\text{ii)} \quad P(x) = (x - (1-i))(x - (1+i))(x - 2) \quad [1]$$

$$= (x^2 - 2x + 2)(x - 2)$$

$\therefore$ by inspection

$$p(x) = (x^2 - 2x + 2)(x - 1) \quad [1]$$



1 mark for correct rays and line

(need to show gradient of -1 or 1, by either marking angle or points)

1 mark for correct region

1 mark for open circle at intersection points at region.

Question 2

a) For every integer n , if n is divisible by 3 then n^2 is divisible by 3

- i) For every integer n , if n^2 is divisible by 3 then n is divisible by 3
ii) For every integer n , if n^2 is not divisible by 3 then n is not divisible by 3
iii) There exists an integer n such that n is divisible by 3 and n^2 is not divisible by 3

b) RTP: n is even if and only if $3n^2 + 8$ is even

Need to prove the following statements

1) if n is even $\Rightarrow 3n^2 + 8$ is even

2) if $3n^2 + 8$ is even $\Rightarrow n$ is even

To prove statement 1, we use a direct proof

(2 marks)

let $n = 2k$, $k \in \mathbb{Z}$

1 mark for
correct method

$$\therefore 3n^2 + 8 = 3(2k)^2 + 8$$

$$= 3(4k^2) + 8$$

$$= 12k^2 + 8$$

$$= 2(6k^2 + 4) \quad (6k^2 + 4 \text{ is an integer})$$

$$\therefore 3n^2 + 8 \text{ is even}$$

2 marks for
correct proof

Now to prove statement 2, we can prove the contrapositive. (2 marks)

if n is not even then $3n^2 + 8$ is not even

1 mark for
correct method

let $n = 2m + 1$ where $m \in \mathbb{Z}$

2 mark for
correct proof

$$3(2m+1)^2 + 8$$

$$= 3(4m^2 + 4m + 1) + 8$$

$$= 12m^2 + 12m + 3 + 8$$

$$= 12m^2 + 12m + 11$$

$$= 2(6m^2 + 6m + 5) + 1$$

$$= 2M + 1 \quad \text{where } M \text{ is an integer}$$

$$\therefore 3n^2 + 8 \text{ is not even}$$

$$\therefore n \text{ is even iff } 3n^2 + 8 \text{ is even}$$

c) Suppose that $2x^3 + 6x + 1 = 0$ has integer solutions, i.e. x is an integer

3 marks

Now $2x^3 + 6x + 1 = 0$

- full correct proof

$$\therefore 1 = -2x^3 - 6x$$

2 marks

$$= 2(-x^3 - 3)$$

as x is an integer then $-x^3 - 3$ is also an integer. ~

$$\therefore 2(-x^3 - 3) \text{ is even}$$

as 1 is not even then a contradiction occurs

$\therefore 2x^3 + 6x + 1 = 0$ has no integer solutions.

Question 3

$$\operatorname{Re}\left(\frac{z-1}{z-i}\right) = 0$$

$$\text{let } z = x+iy$$

$$\therefore \frac{z-1}{z-i} = \frac{x+iy-1}{x+iy-i}$$

$$= \frac{(x-1)+iy}{x+i(y-1)}$$

$$= \frac{[(x-1)+iy][x-i(y-1)]}{x^2-(y-1)^2}$$

$$= \frac{x(x-1) - i(x-1)(y-1) + ixy + y(y-1)}{x^2-(y-1)^2}$$

$$\therefore \operatorname{Re}\left(\frac{z-1}{z-i}\right) = \frac{x(x-1) + y(y-1)}{x^2-(y-1)^2}$$

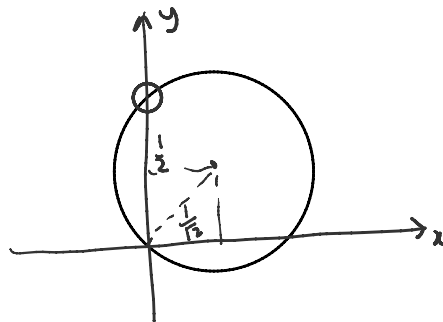
$$\therefore \operatorname{Re}\left(\frac{z-1}{z-i}\right) = 0 \Rightarrow x(x-1) + y(y-1) = 0$$

$$x^2 - x + y^2 - y = 0$$

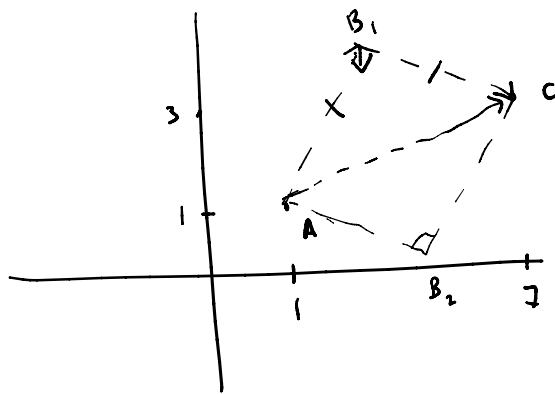
$$\left(x - \frac{1}{2}\right)^2 + \left(y - \frac{1}{2}\right)^2 = \frac{1}{2}$$

$$\therefore \text{circle centre } \left(\frac{1}{2}, \frac{1}{2}\right) \text{ radius } \frac{1}{\sqrt{2}}$$

$$\text{But } z \neq i$$



b)



$$\vec{OB_1} = \vec{OA} + \vec{AB_1}$$

$$\text{Now } B_1C = \vec{AB_1} \times -i$$

$$\vec{AC} = \vec{AB} + \vec{B_1C}$$

$$(7+3i) - (1+i) = \vec{AB_1} - \vec{AB_1} \cdot i$$

$$= \vec{AB_1} (1-i)$$

$$6+2i = \vec{AB_1} (1-i)$$

$$\therefore \vec{AB_1} = \frac{6+2i}{1-i} \times \frac{1+i}{1+i}$$

$$= \frac{6+8i-2}{2}$$

$$= \frac{4+8i}{2}$$

$$= 2+4i$$

$$\therefore \vec{OB_1} = (1+i) + (2+4i)$$

$$= 3+5i$$

$$\text{Now } \vec{OB_2} = \vec{OA} + \vec{AB_2}$$

$$= \vec{OA} + \vec{AB_1} \times -i$$

$$= (1+i) + (2+4i) \cdot -i$$

$$= 1+i - 2i + 4$$

$$= 5-i$$

$$\therefore z_2 = 3+5i \text{ or } 5-i$$

c) $z^7 = 1$

i) let $z = \cos \theta + i \sin \theta$

$$\therefore z^7 = 1$$

$$z^7 = \cos(2k\pi) + i \sin(2k\pi), k \in \mathbb{Z}$$

$$z = [\cos(2k\pi) + i \sin(2k\pi)]^{\frac{1}{7}}$$

$$= \cos\left(\frac{2k\pi}{7}\right) + i \sin\left(\frac{2k\pi}{7}\right) \text{ by De Moivre's Thm, for } k = 1, 2, 3, 4, 5, 6, 7$$

$$\therefore \alpha_k = \cos\left(\frac{2k\pi}{7}\right) + i \sin\left(\frac{2k\pi}{7}\right)$$

ii)

as α_k are the roots of $z^7 = 1$ i.e. $z^7 - 1 = 0$

$$\text{and } z^7 - 1 = (z-1)(z^6 + z^5 + z^4 + z^3 + z^2 + z + 1)$$

then α_k are the roots of $(z-1)(z^6 + z^5 + z^4 + z^3 + z^2 + z + 1)$

as $\alpha_7 = 1$ then α_7 is the root of $z-1$

$\therefore \alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5, \alpha_6$ are the roots of

$$z^6 + z^5 + z^4 + z^3 + z^2 + z + 1 = 0$$

$$\therefore z^6 + z^5 + z^4 + z^3 + z^2 + z + 1 = (z - \alpha_1)(z - \alpha_2) \dots (z - \alpha_6)$$

$$\text{let } z = 1$$

$$\therefore 1 + 1 + 1 + 1 + 1 + 1 + 1 = (1 - \alpha_1)(1 - \alpha_2) \dots (1 - \alpha_6)$$

$$\therefore (1 - \alpha_1)(1 - \alpha_2) \dots (1 - \alpha_6) = 7$$

iii)

$$1 - \alpha_k = 1 - \left(\cos\left(\frac{2k\pi}{7}\right) + i\sin\left(\frac{2k\pi}{7}\right) \right)$$

$$= 1 - \left(1 - 2\sin^2\left(\frac{k\pi}{7}\right) + 2i\sin\left(\frac{k\pi}{7}\right)\cos\left(\frac{k\pi}{7}\right) \right) \quad \text{using double angle identity}$$

$$= 2\sin^2\left(\frac{k\pi}{7}\right) - 2i\sin\left(\frac{k\pi}{7}\right)\cos\left(\frac{k\pi}{7}\right)$$

$$= 2\sin\left(\frac{k\pi}{7}\right) \left[\sin\left(\frac{k\pi}{7}\right) - i\cos\left(\frac{k\pi}{7}\right) \right]$$

$$= 2\sin\left(\frac{k\pi}{7}\right) \left[\cos\left(\frac{\pi}{2} - \frac{k\pi}{7}\right) - i\sin\left(\frac{\pi}{2} - \frac{k\pi}{7}\right) \right]$$

$$= 2\sin\left(\frac{k\pi}{7}\right) \left[\cos\left(\frac{7\pi - 2k\pi}{14}\right) - i\sin\left(\frac{7\pi - 2k\pi}{14}\right) \right]$$

$$= 2\sin\left(\frac{k\pi}{7}\right) \left(\cos\left(\frac{\pi}{14}(7-2k)\right) - i\sin\left(\frac{\pi}{14}(7-2k)\right) \right)$$

$$= 2\sin\left(\frac{k\pi}{7}\right) \left(\cos\left(-\frac{\pi}{14}(2k-7)\right) - i\sin\left(-\frac{\pi}{14}(2k-7)\right) \right)$$

$$= 2\sin\left(\frac{k\pi}{7}\right) \left(\cos\left(\frac{\pi}{14}(2k-7)\right) + i\sin\left(\frac{\pi}{14}(2k-7)\right) \right)$$

$$\text{as } \cos(-\theta) = \cos\theta$$

$$\sin(-\theta) = -\sin\theta$$

iv) from (iii)

$$1 - a_1 = 2 \sin \frac{\pi}{7} \left(\cos \left(\frac{-5\pi}{14} \right) + i \sin \left(\frac{-5\pi}{14} \right) \right)$$

$$\therefore |1 - a_1| = \left| 2 \sin \frac{\pi}{7} \left(\cos \left(\frac{-5\pi}{14} \right) + i \sin \left(\frac{-5\pi}{14} \right) \right) \right|$$

$$= \left| 2 \sin \frac{\pi}{7} \right| \left| \cos \left(\frac{-5\pi}{14} \right) + i \sin \left(\frac{-5\pi}{14} \right) \right|$$

$$= 2 \sin \frac{\pi}{7} \quad \text{as} \quad \sin \frac{k\pi}{7} > 0 \quad \text{for} \quad k = 1, 2, 3, 4, 5, 6.$$

$$\text{and} \quad \left| \cos \frac{\pi}{14} (2k-7) + i \sin \left(\frac{\pi}{14} (2k-7) \right) \right| = 1$$

$$\text{Similarly} \quad |1 - a_2| = 2 \sin \frac{2\pi}{7} \quad \text{etc}$$

$$\therefore |1 - a_1| \cdot |1 - a_2| \cdot \dots \cdot |1 - a_6| = 2 \left(\sin \frac{\pi}{7} \right) \cdot 2 \left(\sin \frac{2\pi}{7} \right) \dots \left(2 \sin \frac{6\pi}{7} \right)$$

$$\left| (1 - a_1)(1 - a_2) \dots (1 - a_6) \right| = 2^6 \left(\sin \frac{\pi}{7} \right) \left(\sin \frac{2\pi}{7} \right) \dots \sin \frac{6\pi}{7}$$

$$|7| = 2^6 \left(\sin \frac{\pi}{7} \right) \left(\sin \frac{2\pi}{7} \right) \dots \sin \frac{6\pi}{7} \quad (\text{From (ii)})$$

$$\therefore \left(\sin \frac{\pi}{7} \right) \left(\sin \frac{2\pi}{7} \right) \dots \left(\sin \frac{6\pi}{7} \right) = \frac{7}{2^6}$$